



The Ratio of N-Methyl-2-Pyrrolidone (NMP) To Other Components In Lithium Ion Half Cells

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Background

Lithium Ion batteries have become vital to our day to day lives. Producing coin cells in the lab allows researchers to work with the batteries in their prime rather than ones that have degraded during transit, but companies like Duracell and Energizer keep their ratio of ingredients a secret.



Introduction

The goal of this research is to determine the ratio of N-methyl-2-pyrrolidone(NMP) to all other components in lithium manganese oxide (LMO) and lithium titanate oxide (LTO) half cells.

LMO is the positively charged cathode and LTO is the negatively charged anode.

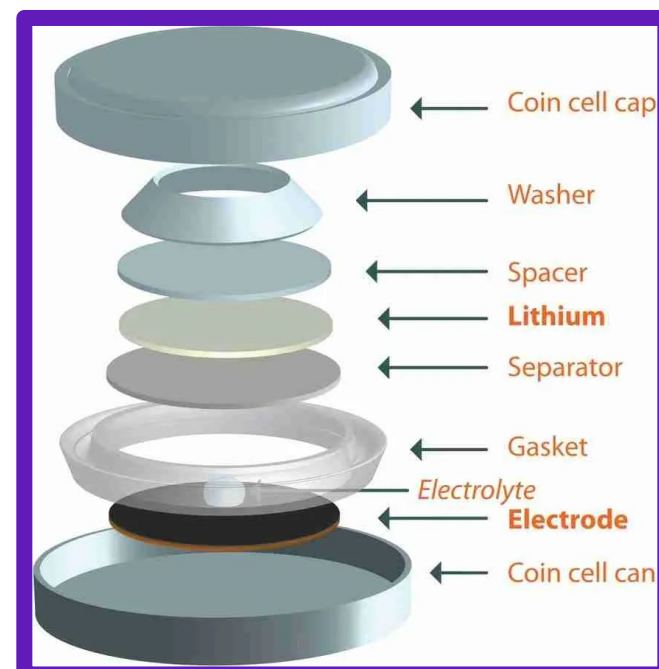


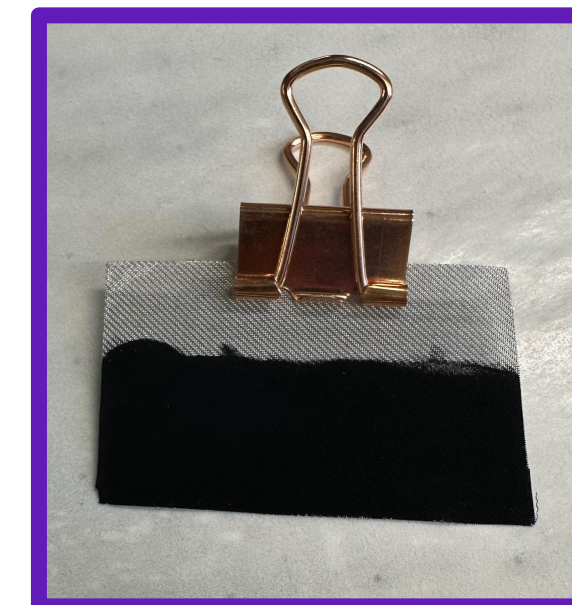
Figure 1



Figure 2

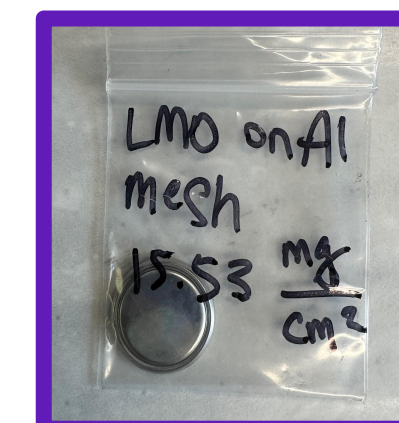
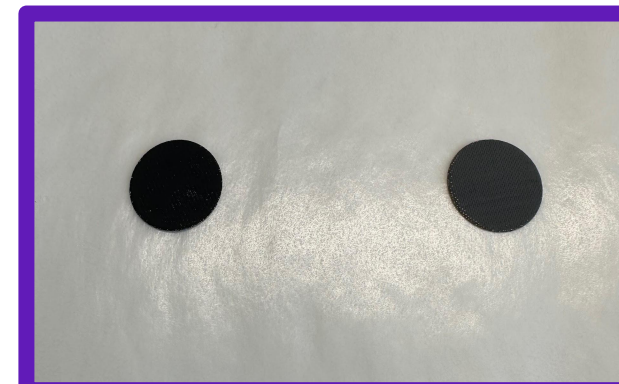
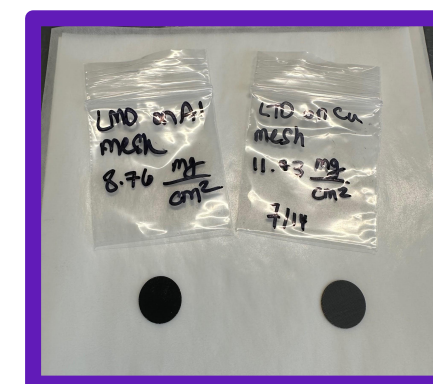
Materials and Methods

Adjusting the quantity of the NMP and the volume of slurry optimizes the coating and loading density of the final electrode. For LTO, the current collector is copper mesh and for LMO, it is aluminium mesh. Varying the volume of slurry also affects the loading density,



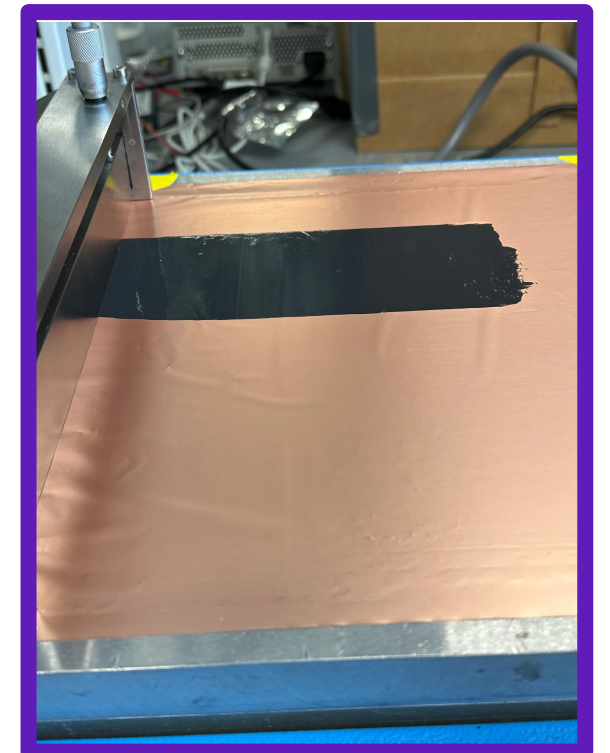
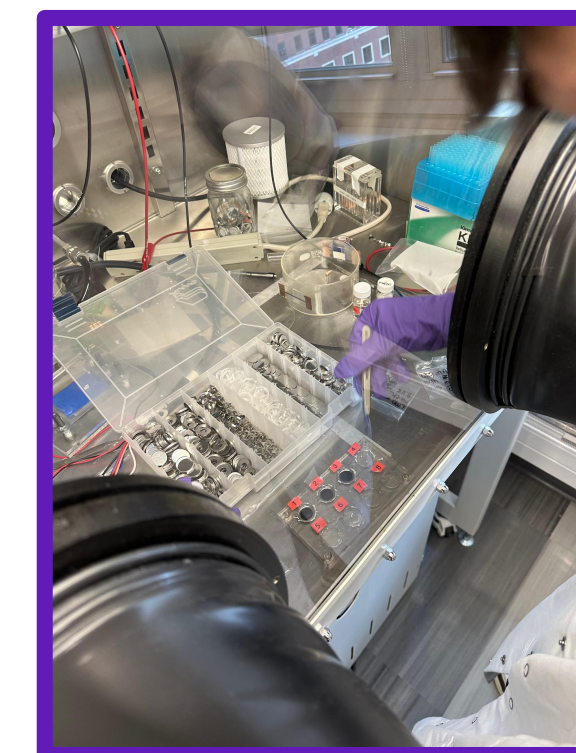
Results

For LTO coating, a ratio of 1.7 NMP to all other components by mass is optimal. For LMO coating, a ratio of 1.6 NMP to all other components is best. Roughly 0.25g of slurry for each 15x15mm area of mesh, is best for enhancing the electrodes



Discussion and Conclusion

Finding the ratio of NMP to all other components is critical for stable, relevant baseline performance. In our case, the electrodes we have built will be added to a window cell for red light induced fast charging and the ratios will be used to make foil coatings for similar research on pouch cells. Despite the fact that this is preliminary research, these ratios are vital for future work as industry controls access to them.



References

1. "CR2032/2025/2016 CR2330/2430 Coin Cell Case Sets, SS304." Landt Instruments, 12 Apr. 2022, www.landtinst.com/cr2032-cr2025-cr2016-cr2430-coin-cell-case-sets-ss304/. 2. STARTECHdotCOM. "CR2032 Lithium Thick Coin Cell Battery - Computer Batteries." StarTech.Com, www.startech.com/en-us/computer-parts/rv2032b. Accessed 8 Aug. 2023.

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